

Connectorized Optical I/O Chiplet with V-groove for AI and High Performance Computing

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Abstract: This paper presents connectorized in-package optical I/O chiplets with V-groove for passive fiber attach, enabling robust and scalable connectivity solutions for AI and high-performance compute. We demonstrate a process for ensuring known good chiplets using Ayar Labs' TeraPHY™ optical I/O chiplet.

1. Introduction

The in-package optical I/O chiplet is an emerging technology aimed at addressing the computing bandwidth challenges in AI and high-performance computing (HPC) applications. However, integration of these chiplets into a large compute multi-chip package (MCP) presents significant challenges for the industry. This paper summarizes recent advancements in packaging options for in-package optical I/O chiplets and proposes one solution: a connectorized optical I/O chiplet with V-groove technology. The TeraPHY optical I/O chip from Ayar Labs [1] is utilized to demonstrate this packaging approach. The chip's V-groove enables passive fiber attachment, making it scalable for high-volume manufacturing (HVM). The chiplet is paired with a multi-channel optical connector with lens array (MOLA), which is specifically designed and optimized for integration with the TeraPHY optical chiplet in an MCP. The connector's compact design ensures compatibility with HVM processes. To mitigate the risk of yield loss in optical assembly, which could impact the overall cost of compute, memory, or fabric MCPs, it is crucial that the optical chiplet be tested for both electrical and optical functionality before assembly into the final package. This paper provides a description of the process flow required to enable known good connectorized chiplet (KGCC) for such applications.

2. Optical Interconnect in Silicon Photonics Package

The optical package architecture for photonic integrated circuits (PICs) consists of a first-level optical interface, a second-level optical interface [2], and the optical interconnect between them, as summarized in Table 1. The optical interconnect serves several critical functions:

- It provides the optical connection between the PIC and the external fiber communication system.
- It acts as a stress buffer, mitigating the mechanical stress on the PIC caused by external forces such as the plugging and unplugging of fiber components, or forces transferred from the external fiber system.

Table 1. Components in optical package architecture

Component	Options	Note
First-level optical interface	permanent bond	Usually requires UV curable epoxy for bonding
	detachable	Teramont's TeraVERSE photonic bump is one emerging option [3]
Optical interconnect	fiber bundle	Traditional approach, mature, higher cost
	glass/Si waveguide	Rigid. Intel's detachable connector is one emerging option [4]
	polymer waveguide	Early phase. Insertion loss (IL) needs to be improved before adoption.
Second-level optical interface	MT ferrule	Most available. Large size. Can't survive standard solder reflow. High IL variation. Difficult to plug/unplug.
	next gen connector	New standard is needed.

The optical package architecture presented in this paper features a permanent bond for the first-level optical interface, a fiber bundle as the optical interconnect, and a second-level optical interface, with MOLA being a potential option for next-generation connectors, as shown in Figure 1 (b). The fiber bundle is proposed due to its maturity as a technology, making it suitable for early product introduction to the market. The first-level optical interface employs edge coupling with fibers positioned in a V-groove. The V-groove as shown in Figure 1 (a) is fabricated with GlobalFoundries Fotonix platform, for which Ayar Labs has multiple years' experience to mature the V-groove fiber attach process.

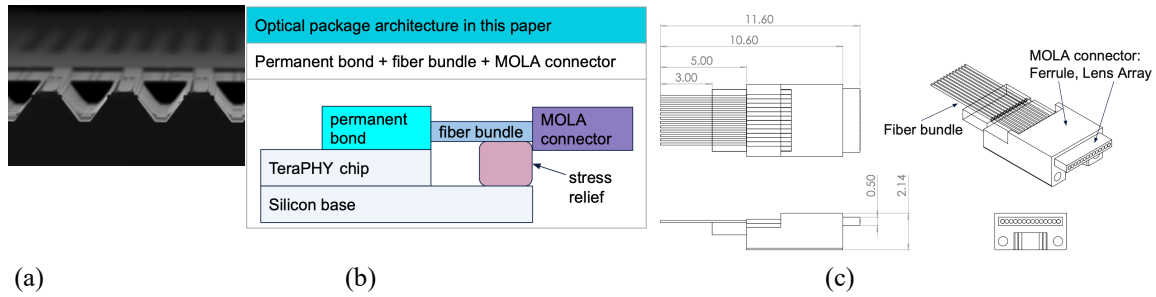


Figure 1: Optical package architecture on Silicon Photonics chip with V-groove, and structure of MOLA FAU

3. Design and Performance of MOLA Connector

The structure of the MOLA fiber array unit (FAU) is shown in Figure 1 (c). The current design features a 12-channel fiber bundle that can be scaled up to a 24-channel bundle if required. On the left side of the FAU, the cleaved bare fiber fits into the V-groove on the silicon photonics die. The fiber bundle is assembled into a customized MOLA ferrule, which is compatible with metal pin insertion. The female MOLA ferrule, depicted in Figure 1 (c), will serve as the package connector. A silicon lens array is attached to the MOLA ferrule using active alignment, with future development aimed at enabling passive alignment.

Figure 2 presents the simulated performance of the MOLA connector. The expanded beam within the MOLA design helps to relax the translational shift tolerance when mating two MOLA connectors, although it tightens the angular tilt tolerance. Extensive engineering efforts were made to optimize the beam size after the lens. With this optimization, the angular tolerance for a 0.2 dB IL impact is ± 0.2 degrees, and the translational tolerance in the plane perpendicular to the optical axis for a 0.2 dB IL impact is 6 μm . For a 0.2 dB IL impact, the distance tolerance is ± 600 μm , allowing for easy plug-and-play functionality of the MOLA connector.

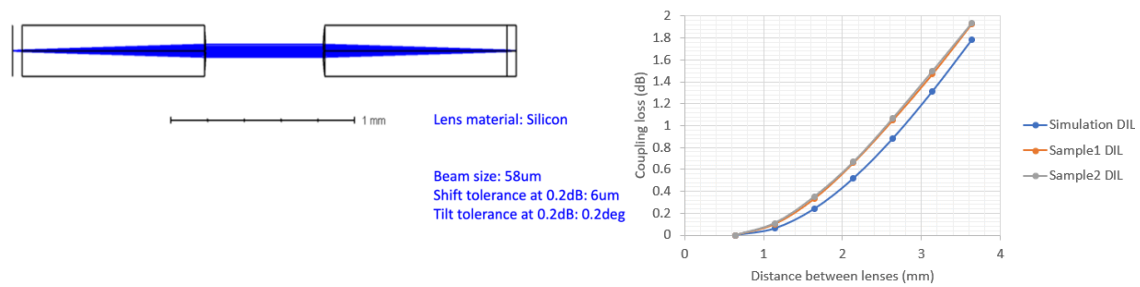


Figure 2: Coupling loss vs. shift and tilt and distance between lenses

Figure 3 shows the repeatability measurements of MOLA connectors after 10 plug-and-unplug cycles. The standard deviation of the repeatability test is less than 0.1 dB, based on early proof-of-concept samples. Further improvements are anticipated as more samples are developed and tested.

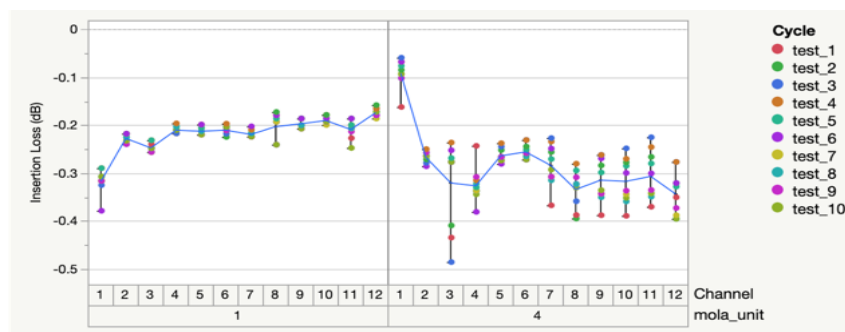


Figure 3: Repeatability test of MOLA plug and unplug

For free-space beam propagation between the silicon lenses, dust protection is a critical consideration. In this paper, we applied a silicone dust seal around the lens arrays to prevent dust particles from entering the space between the

lenses during operation. During manufacturing and assembly, a silicone dust cap is used to protect the lens surface, and an IPA wipe is expected to be sufficient for cleaning the lens surface.

4. Process Flow for Connectorized Optical I/O Chiplet

The known good die (KGD) concept is well established in the semiconductor packaging industry, ensuring that each die is tested and confirmed to be functional before being assembled into a multi-chip package. With the MOLA connector, we introduce the known good connectorized chip (KGCC) concept, where the photonic chip is connectorized with a fiber attachment process and tested before being integrated into a multi-chip package containing costly components like CPUs, GPUs, or AI chips. The benefits of KGCC are outlined below:

- Mitigate fiber attachment yield loss: By moving the fiber attachment process to an earlier stage in the packaging assembly flow, potential yield losses related to fiber attachment are minimized. Avoiding the Combined Yield Loss at the MCP level, where several Optical I/O would be added to the Assembly.
- Eliminate package substrate design ‘cut-outs’: This approach maximizes the routing “real estate” on the substrate by removing the necessity for cut-outs in the package design. In traditional processes where fiber attachment occurs after flip-chip assembly, cut-outs in the substrate are required to allow the optical components to access the chip coupler on the same side as the bumping.

The process flow for the known good connectorized chip (KGCC) is shown in Figure 4. After wafer fabrication, the wafer undergoes Cu pillar bumping, followed by both electrical and optical wafer sorting. The die preparation process then singulates the wafer into individual dies, which are packed into shipping media, either waffle packs or tape-and-reel. This completes the KGD process. In the next step, the MOLA FAU is attached to the chip and tested both electrically and optically, resulting in the KGCC. The KGCC is then flip-chipped onto a package substrate with underfill. An integrated heat spreader may be added, depending on customer preference. Finally, the assembled package is screened electrically and optically as the last step.

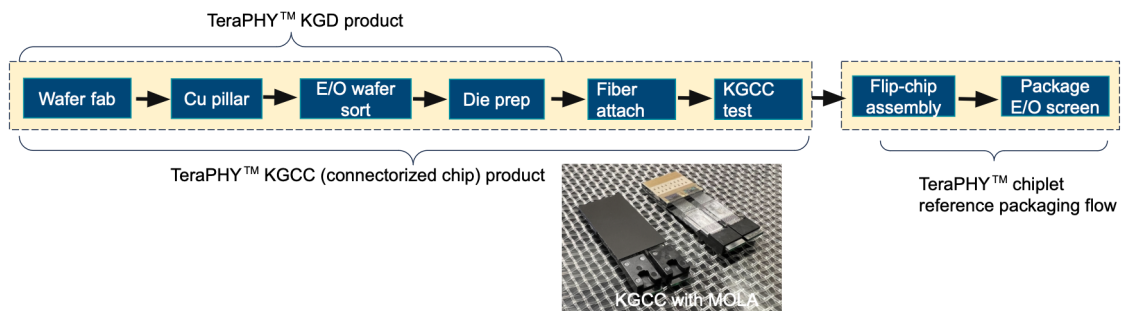


Figure 4: Process flow for KGCC

5. Conclusion

The KGCC with a MOLA connector was proposed as one product offering for an optical I/O chiplet. The MOLA connector performance data is promising from the proof-of-concept samples, demonstrating relaxed tolerance in translation shift and stable plug and unplug repeatability data. The KGCC process flow can help to mitigate the risk of fiber attach yield loss and eliminate the cutout in the package substrate normally required for the photonics chip with chip coupler on the bumping side.

6. Acknowledgement

We acknowledge the support from GlobalFoundries, Browave, Powertech Technology, and Fabrinet for sample fabrication and assembly.

7. References

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